TECHNICAL RESEARCH SUMMARY | WITH PRELIMINARY DATA

Current-Based Diagnosis of Inverter Open-Circuit and Sensor Gain Faults in PMSM Drives

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Research objective and method

This study uses three-phase currents to diagnose inverter open-circuit faults and single-phase current-sensor gain faults in a permanent-magnet synchronous motor (PMSM) drive. The fault space includes six single-switch and fifteen double-switch configurations. Sensor gain faults affect one measured phase while the other two sensor channels remain normal. The current pipeline applies **pointwise L2 normalization** to the three-phase vector, then extracts **positive- and negative-half-wave current features** for fault discrimination. A gain error in one phase changes the shared L2 denominator, so normalized features in all three phases can be affected.

Open-circuit evaluation: one saved, fixed-split study

The reported test subset contains **66 simulation records**: 18 single-switch, 45 double-switch and 3 normal records, at **1000 rpm and 10 N m**, with fault-injection angles of **90, 210 and 330 degrees**. The Angle9 L2-feature model was retained; a confidence threshold of **0.90** was selected on validation data and evaluated on test with the same **1 ms confirmation rule**.

Test measure	Observed result
Single-switch locks within 0.5 electrical cycle	2 / 18 (11.1%)
Double-switch locks within 0.75 electrical cycle	42 / 45 (93.3%)
Median lock delay: single / double switch	12.803 / 7.614 ms
Wrong final fault locks / normal false locks	0 / 63 fault records; 0 / 3 normal records

All 63 fault records eventually locked correctly in this saved run, but the **single-switch timing target remains unmet in most cases**. These are record-level lock results for this operating condition, not a claim of accuracy across all conditions.

Sensor gain drift: exploratory perturbation replay

A separate saved sensor-diagnosis prototype was tested on **108 correlated gain-perturbation cases** derived from **3 normal parent records**: 3 affected phases, 6 gain factors (0.70, 0.80, 0.925, 1.075, 1.20, 1.30), and 2 stored ramp settings. It recorded **108/108 correct locks**, no wrong locks and no open-switch false locks on those sensor cases; median lock delay was **9.073 ms**. These are synthetic, correlated stress tests from a separate prototype, not 108 independent experiments or a validated final joint model.

Current research focus

Improve early single-switch discrimination while controlling wrong locks, and validate single-phase sensor gain drift diagnosis using the L2-normalized half-wave pipeline on more independent records and operating conditions. The data above are preliminary simulation results; hardware validation and final generalization performance remain open.

Data provenance: Open-circuit counts and delays recalculated from TEST_RECORDS_DEFAULT_VS_SELECTED.csv (Angle9, test, validation_selected), cross-checked against TEST_SUMMARY.csv. Sensor figures recalculated from GAIN_test_records.csv and checked against GAIN_parent_summary.csv. Results from the two saved studies are reported separately.